

End of Life Care in the ED



Section I: Scenario Demographics

Scenario Title:	End of Life Care in the ED
Date of Development:	(08/12/2016) Revised (31/03/2018)
Target Learning Group:	☐ Juniors (PGY 1-2) ☐ Seniors (PGY $\geq$ 3) ☒ All Groups

Section II: Scenario Developers

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Section III: Curriculum Integration

Objectives	
Educational Goal:	Expose learners to the assessment and medical management of a palliative patient who presents to the emergency department with severe shortness of breath.
CRM Objectives:	1) <i>Communicate</i> effectively with the patient, substitute decision maker and health care team during a critical clinical situation. 2) Provide <i>leadership</i> to a team caring for a patient in extremis whose goals of care are not full resuscitation.
Medical Objectives:	1) Verbalize and work through the causes of acute respiratory distress. 2) Provide rapid symptom control treatment at the end of life. 3) Communicate with patient and or substitute decision maker about the likely medical outcomes of various critical care interventions.

Case Summary: Brief Summary of Case Progression and Major Events

A 72-year old male with small cell lung cancer and bony metastases presents with acute shortness of breath. Curative treatment has been stopped and palliative care assessment is pending. He is on home oxygen and has come to the ED as his symptoms could not be controlled at home.

The patient initially improves with oxygen and pain control. He is too confused to engage in discussion about advanced directives. No previous advanced directives or level of care have been documented but, Cindy, the patient's daughter is available to act as decision maker. She will have a number of questions about her father's care.

The patient's respiratory status will deteriorate. Cindy will confirm her father's wish for comfort measures, to be started by the treating team.

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Section IV: Scenario Script

A. Scenario Cast & Realism			
Patient:	☐ Computerized Mannequin	Realism:	☒ Conceptual
	☒ Mannequin		☐ Physical
	☐ Standardized Patient		☒ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A
		<i>Select most important dimension(s)</i>	
Confederates			
Brief Description of Role			
Cindy	Daughter – Only child and substitute decision maker (SDM)/Power of Attorney (POA). Cindy and her father know the lung cancer has progressed and their oncologist has referred them to palliative care but they don't have any prior experience with this and don't know what it means. (Her medical literacy is below average.) Her mother (patient's wife) passed away in ICU 2 years ago after multiple admissions for CHF. The treatment was challenging and Mr. Clarke would not want this for himself. Acts as SDM, wants interventions if "it will help". On further questioning, she reveals that her father has said recently that he is tired of all the pain and does not want any painful procedures or "heroic measures". Cindy is conflicted between wanting her father to live and following his wishes to avoid further interventions. <i>May be present at bedside or available on phone, depending on sim team availability.</i>		
B. Required Monitors			
☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line	
☒ NIBP Cuff	☒ Defibrillator Pads	☒ Capnography	
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:	
C. Required Equipment			
☒ Gloves	☒ Nasal Prongs	☐ Scalpel	
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit	
☐ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit	
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit	
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit	
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit	
☐ Blood Products	☒ ET Tubes	☐ Other:	
☐ Intraosseous Set-up	☐ LMA	☐ Other:	
D. Moulage			
Elderly man wearing PJs.			
E. Approximate Timing			
Set-Up:	5min	Scenario:	12 min
		Debriefing:	15 min

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Section V: Patient Data and Baseline State

A. Clinical Vignette: To Read Aloud at Beginning of Case

"Mr. Clarke is a 72 year old man from home with acute shortness of breath. He is in the resuscitation room. He has a history of lung cancer and is on 2L home oxygen. His daughter Cindy called 911 because he has been getting worse since this morning. He just finished a course of antibiotics for presumed pneumonia. He is on hydromorph contin and prochlorperazine. No allergies. Here is his most recent oncology clinic note."

B. Patient Profile and History

Patient Name: Matt Clarke		Age: 72	Wt: 60 kg
Gender: ☒ M ☐ F		Code Status: unknown	
Chief Complaint: Acute Severe Shortness of breath			
History of Presenting Illness: Since this morning patient has become more short of breath. His daughter found him struggling to breathe, cyanotic and not able to speak more than 1-2 words. He is acutely confused. She called 911, and EMS brought him to the hospital. He is also in significant pain from his metastases to his R femur and lumbar spine (missed pain meds at home).			
Past Medical Hx:	▪ SCLC for 1.5 yrs, progression despite chemo and radiation ▪ Mets to bones – severe back and hip pain ▪ Home O2 (2L NP) ▪ Palliative Care consult pending	Meds:	▪ Hydromorph Contin 9 mg BID ▪ Hydromorph 2 mg breakthrough ▪ Prochlorperazine 5mg PRN ▪ Levofloxacin 750 mg – finished 5 days ago for pneumonia

Allergies: None

Social History: Lives at home with his daughter Cindy and her family

Family History: None

Review of Systems:	CNS:	Alert but confused, follows command	
	HEENT:	Lips look cyanotic	
	CVS:	Normal	
	RESP:	Resp distress, using accessory muscles, speaks 1-2 words	
	GI:	Normal	
	GU:	Normal	
	MSK:	Complains of leg and hip pain	INT: Normal

C. Baseline Simulator State and Physical Exam

☐ No Monitor Display		☒ Monitor On, no data displayed		☐ Monitor on Standard Display	
HR: 120/min		BP: 130/80		RR: 30/min	
Rhythm: Sinus		T: 36.8°C		Glucose:7 mmol/L	
				O ₂ SAT:84 % (4L NP)	
				GCS: 14 (E4 V4 M6)	
General Status: Increased work of breathing, alert but confused					
CNS:	Alert but confused.				
HEENT:	Lips are cyanotic				
CVS:	S1S2 no murmur, JVP is normal, no leg edema				
RESP:	Decreased on right side, increased work of breathing, 1-2 works sentences due to dyspnea				
ABDO:	Soft, tender				
GU:	Normal				
MSK:	Tenderness on palpation of right leg and hip			SKIN:	



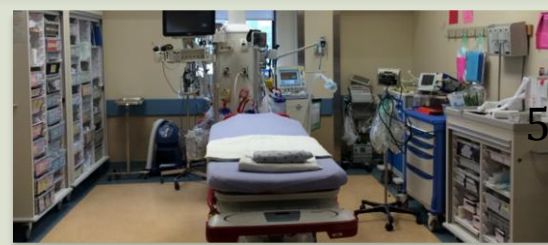
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Section VI: Scenario Progression

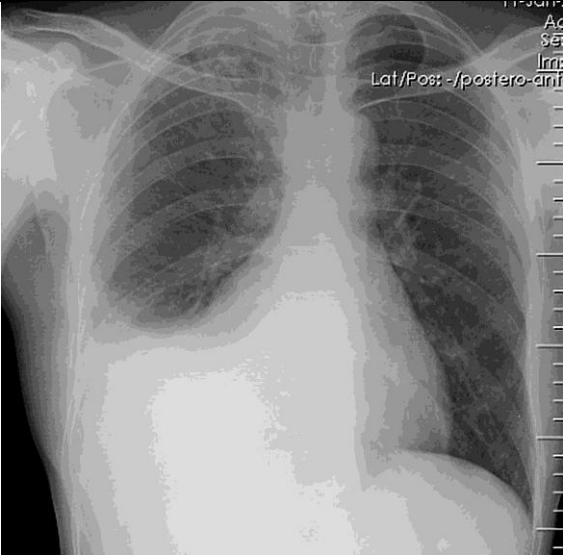
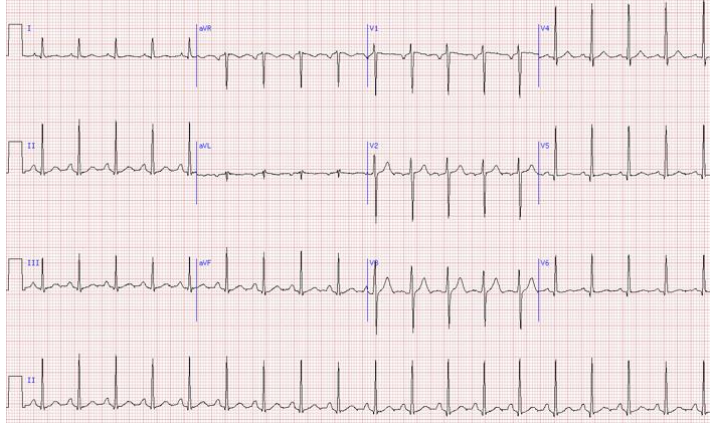
Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: Sinus tach HR: 120/min BP: 130/80 RR: 30/min O ₂ SAT:84 % (2L NP) T: 36.8°C	Sitting upright Patient complains of hip and back pain. "My back...my hip. Need meds." "It hurts." "Can't breathe." All other responses: "don't know" and "ask Cindy."	Learner Actions ☐ History & physical ☐ IV, blood work, EKG ☐ Saline bolus ☐ Glucose check ☐ CXR ☐ Call RT ☐ Apply O2 ☐ Team member to read clinic note	Modifiers <i>Changes to patient condition based on learner action</i> -Ask patient about his goals of care, he will say "ask Cindy" Triggers <i>For progression to next state</i> -Read clinic note or 5 mins into the case → 2. Stabilization
2. Stabilization Rhythm: Sinus tach HR: 110/min BP: 140/90 RR: 24/min O ₂ SAT:94% (NRB) T: 36.8°C		Learner Actions ☐ Try to find code status documentation ☐ POCUS for PCE, CHF, PNA and PTX ☐ Treat bone pain ☐ Call SDM	Modifiers -Pain meds given → HR 100 -If attempt intubation, daughter to arrive/phone asking for MD Triggers -Call POA or attempt intubation → 3. Goals of Care Discussion
3. Goals of Care Discussion Rhythm: Sinus tach HR: 100/min BP: 140/90 RR: 24/min O ₂ SAT:92 % (NRM)	Cindy arrives or on the phone. Asks about current status. Wants to know prognosis and options. Wants to know your recommendation.	Learner Actions ☐ Goals of care discussion with POA ☐ Call Oncology ☐ Call Palliative care	Modifiers -Oncology/Palliative care will agree to see in ED in 30 min Triggers -All actions complete or 7 min into the case → 4. Worsening resp status
4. Worsening Respiratory Status Rhythm: Sinus tach HR: 120/min BP: 120/80 RR: 35/min O ₂ SAT:86 % (NRB)	Patient complains of ongoing shortness of breath. Cindy declines all interventions, wants father to be "comfortable".	Learner Actions ☐ Ongoing goals of care discussion w/ POA ☐ Discuss Intubation ☐ Discuss BiPAP ☐ Discuss thoracentesis ☐ Recommend comfort measures	Modifiers Triggers -GoC discussion complete → 5. Resolution
5. Resolution Monitors shut off		Learner Actions ☐ Shut off monitors ☐ Order narcotic ☐ Order sedative ☐ Call SW/chaplain	Triggers Oncology arrives → End of Case





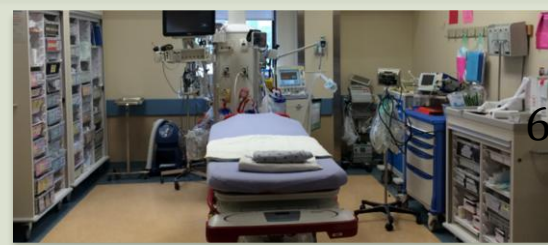
Section VII: Supporting Documents, Laboratory Results & Multimedia

Laboratory Results					
Critical VBG	pH: 7.22	PCO ₂ : 65	PO ₂ : 45	HCO ₃ :28	Lactate:2.5

Images (ECGs, CXRs, etc.)	
 https://radiopaedia.org/cases/pleural-effusion-7	EKG: Sinus tachycardia  http://www.thecrashcart.org/case-2-post-partum-palpitations/

Ultrasound Video Files (if applicable)	
Ultrasound: Right pleural effusion	Ultrasound: No pericardial effusion

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Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
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Sample Questions for Debriefing	
Medical Knowledge - Describe your differential diagnosis for acute respiratory distress. How does this change in an oncology patient? - What are the bedside diagnostic tools for acute respiratory distress? - What are common end-of-life symptoms that are distressful to patients and families? How do we treat them in the ED? - How do you determine a patient's competence? What is the hierarchy of SDMs?	
CRM Skills - What is your approach to goals of care discussions in the ED? How do you modify this over the phone vs in person? Or if time is short because of the clinical situation? - What language do you use to explain critical care interventions to non-medical family members? - What resources can you use when facing an uncertainty in code status? - What are some tools to help navigate conflict regarding treatment decisions <i>within the medical team</i>?	
Key Moments	
Establishing the patient's significant past medical history	
Determining the patient's level of competence and contacting the substitute decision maker	
Goals of care discussion with SDM	
Transitioning clinical management to comfort measures	

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References

- Marx, J. A., Hockberger, R. S., Walls, R. M., & Adams, J. (2013). *Rosen's emergency medicine: Concepts and clinical practice*. St. Louis: Mosby.
- Helman A. (2015). End of life care in emergency medicine. *EMCases.com*.
<https://emergencymedicinecases.com/end-of-life-care-in-emergency-medicine/>
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